KINGDOM OF SAUDI ARABIA
Ministry of Higher Education
Taibah University
College of Computer Science and Engineering

المملكة العربية السعودية
وزارة التعليم العالي
جامعة طيبة
كلية علوم و هندسة الحاسبات

AI342 Course Project

Image Processing Project

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Section: F3

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1. Introduction

This project presents an image processing framework designed to analyze and process different types of traffic sign images. The dataset includes several traffic signs with different geometric structures such as triangular and circular shapes.

The framework consists of multiple stages including Original Image preparation, Noise Injection, Restoration & Enhancement, Segmentation & Morphology, Metrics Evaluation, and Feature Extraction. Different filtering and enhancement techniques were applied to evaluate their effect on image quality and object extraction performance.

In the final stage, structural features such as Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented objects to support image analysis and classification. The system also supports object classification using PCA mapping and minimum distance classification to predict object classes based on the training reference.

2. Dataset Overview

The dataset consists of five traffic sign images with different structural shapes, including circular and triangular signs, to provide structural diversity for image processing analysis. Circular signs were selected to evaluate shape preservation and boundary smoothness, while triangular signs were chosen to test edge detection and polygonal structure preservation.

Different types of noise were either identified or artificially injected into the images, including Gaussian noise, Salt & Pepper noise, and Exponential noise. Appropriate restoration and enhancement filters were then applied to reduce degradation while preserving the important structural details of each traffic sign. The dataset was further used for segmentation, morphology, feature extraction, and PCA-based classification analysis.

2. Traffic Sign Image Analysis

Image 1 – Triangular Traffic Warning Sign (Aram)

1. Original Image

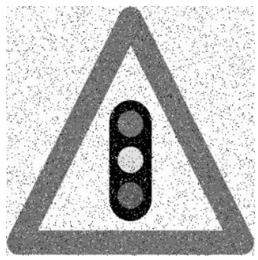
The selected image represents a traffic sign with clear geometric boundaries and distinct structural features. The image was converted into grayscale to simplify image processing operations while preserving the important shape details required for further analysis and classification.

Original Image

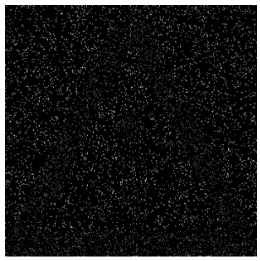
Grayscale Image

2. Noise Injection

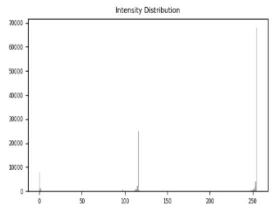
Salt & Pepper noise was injected into the grayscale image to simulate image degradation and evaluate the robustness of the restoration stage. The added noise introduced random black and white intensity variations across the image and slightly affected edge continuity. The Difference Map highlights the pixel changes caused by the noise, while the histogram illustrates the distribution of intensity values after noise injection.



Noisy Image



Difference Map

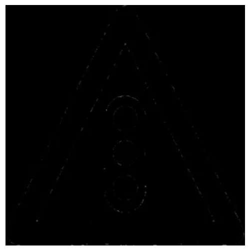


Histogram

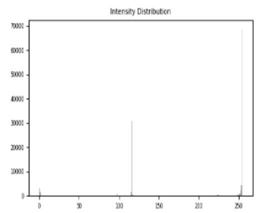
3. Restoration & Enhancement

A Median filter with a kernel size of 5 was applied to reduce Salt & Pepper noise while preserving important structural edges. The restored image showed reduced noise distortion and improved visual clarity compared to the noisy image. The Difference Map and histogram also indicate successful restoration and better intensity distribution after filtering

Restored Image



Difference Map

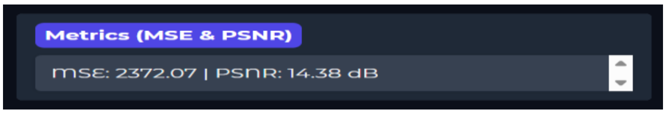


Histogram

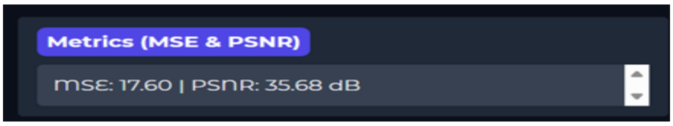
4. Metrics Evaluation

The restored image achieved a significant reduction in MSE and a noticeable increase in PSNR after applying the restoration filter. These results indicate successful noise reduction, improved image quality, and better preservation of important structural details.

Metrics Before Filtering:



Metrics After Filtering:



5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was used to isolate the foreground object from the background. A circular structuring element of size 5 was selected to preserve the outer triangular boundary without introducing distortion.

6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented object. The extracted feature values indicate that the object has polygonal characteristics, which is consistent with the triangular structure of the traffic warning sign.

Feature Vector Results:

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
54650.5	2599.2	0.102	123.615	17	

7. Final Analysis

The final processing pipeline successfully isolated the traffic sign and extracted meaningful geometric features suitable for classification and PCA analysis.

Image 2 – Circular Traffic Sign Image Analysis (Aram)

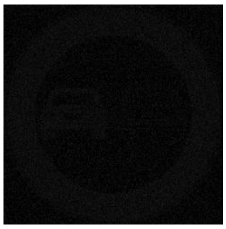
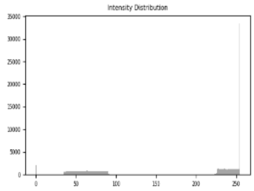
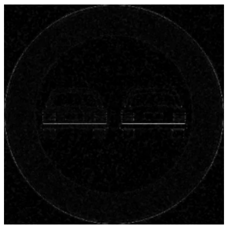
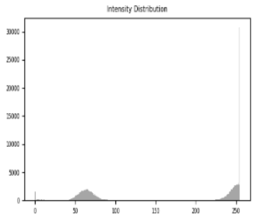
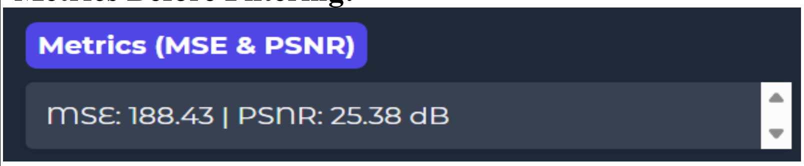
1. Original Image

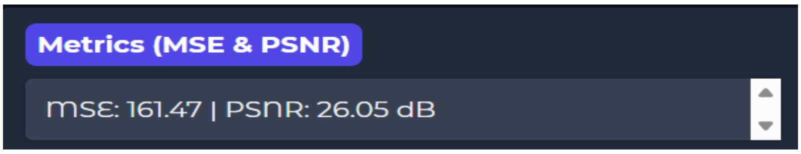
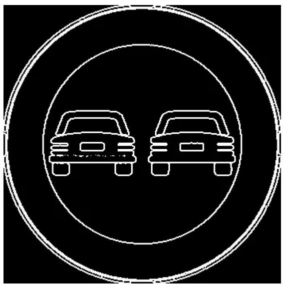
The selected image represents a traffic sign with clear circular boundaries and distinct geometric features. The image was converted into grayscale to simplify image processing operations while preserving the important structural details required for further analysis and classification.

Original Image

Grayscale Image

2. Noise Injection

<table>
<tr><td>Uniform noise was injected into the grayscale image to simulate image degradation and evaluate the robustness of the restoration stage. The added noise introduced random intensity variations across the image and slightly affected the smoothness of the circular boundaries. The Difference Map highlights the pixel changes caused by the noise, while the histogram illustrates the distribution of intensity values after noise injection.</td><td>
Noisy Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">3. Restoration & Enhancement</td></tr>
<tr><td>A Median filter with a kernel size of 3 was applied to reduce Uniform noise while preserving the circular boundaries and important structural details. The restored image showed slight noise reduction and improved visual smoothness compared to the noisy image. The Difference Map and histogram indicate moderate restoration performance after filtering.</td><td>
Restored Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">4. Metrics Evaluation</td></tr>
<tr><td>The restored image achieved a slight reduction in MSE and a small increase in PSNR after applying the restoration filter. These results indicate moderate improvement in image quality and partial noise reduction while preserving important structural features.</td><td colspan="3">Metrics Before Filtering:


Metrics After Filtering:

</td></tr>
<tr><td colspan="4">5. Segmentation & Morphology</td></tr>
<tr><td>Global thresholding with a threshold value of 127 was applied to isolate the circular traffic sign from the background. Boundary extraction using a circular structuring element of size 5 successfully preserved the outer circular contour and internal vehicle structures while maintaining smooth object boundaries.</td><td colspan="3"></td></tr>
</table>

6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented object. The high circularity value confirmed that the object belongs to the circular class, which is consistent with the geometric structure of the traffic sign.

Feature Vector Results:

Area	Perimeter	Circularity	Compactness	Euler	Class
97518	1174.3	0.889	14.141	22	

7. Final Analysis

The final processing pipeline successfully isolated the circular traffic sign and extracted meaningful geometric features suitable for classification and PCA analysis. The segmentation and restoration stages preserved the important circular boundaries and structural characteristics of the object.

Image 3– Slippery Road Traffic Sign (Aram)

1. Original Image

The selected image represents a triangular slippery road warning sign with clear geometric boundaries and internal directional curves. The image was selected because it contains distinct structural features suitable for shape-based analysis and classification.

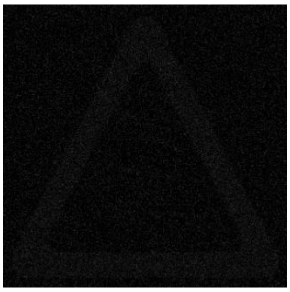
Original Image

Grayscale Image

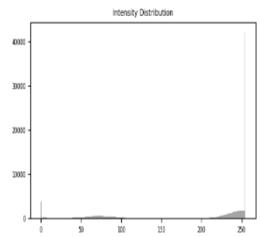
2. Noise Injection

Gaussian noise was injected to simulate random intensity degradation and evaluate the robustness of the restoration stage. The added noise introduced visible intensity variations across the image while partially affecting edge smoothness and structural clarity.

Noisy Image



Difference Map



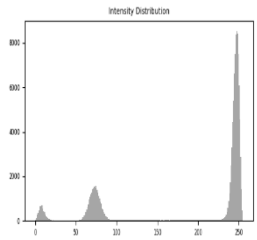
Histogram

3. Restoration & Enhancement

A Mean filter with a kernel size of 3 was applied to reduce Gaussian noise while preserving the main structural edges of the traffic sign. The restored image showed smoother intensity distribution and improved visual clarity with reduced noise distortion.

Restored Image

Difference Map



Histogram

4. Metrics Evaluation

The restoration stage improved image quality by reducing the MSE value from 231.70 to 183.77 and increasing the PSNR value from 24.48 dB to 25.49 dB, indicating successful noise reduction and enhanced restoration performance.

Metrics Before Filtering:

Metrics (MSE & PSNR)

MSE: 231.70 | PSNR: 24.48 dB

Metrics After Filtering:

Metrics (MSE & PSNR)

MSE: 183.77 | PSNR: 25.49 dB

5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was applied to isolate the main structural regions of the traffic sign. Boundary extraction with a circular structuring element of size 5 was used to emphasize the outer triangular edges and improve structural shape representation while preserving important boundary details.

6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented boundary image. The circularity value of 0.607 confirmed that the detected object belongs to a polygonal warning sign rather than a circular class.

Feature Vector Results:

Area	Perimeter	Circularity	Compactness	Euler	Class
61361.5	1127.5	0.607	20.718	5	

7. Final Analysis

The complete processing pipeline successfully enhanced the noisy traffic sign image, extracted meaningful structural boundaries, and generated representative geometric features suitable for image classification and PCA-based analysis.

Image 4 – Triangular Pedestrian Warning Sign (Aram)

1. Original Image

The selected image represents a triangular pedestrian warning sign with clear geometric boundaries and distinct structural features. The image was selected because it contains recognizable shapes suitable for restoration, segmentation, and feature-based analysis.

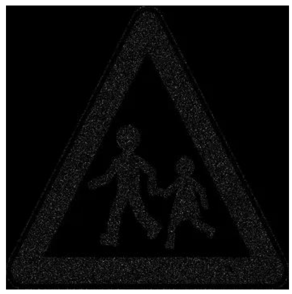
Original Image

Grayscale Image

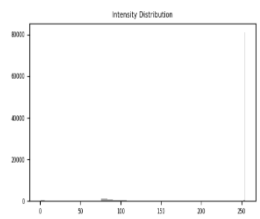
2. Noise Injection

Exponential noise was injected to simulate intensity-based image degradation and evaluate the robustness of the restoration stage. The added noise introduced noticeable intensity variations while partially affecting edge clarity and structural details across the image.

Noisy Image



Difference Map



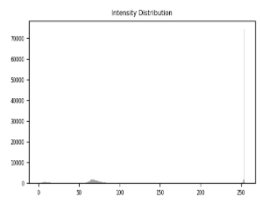
Histogram

3. Restoration & Enhancement

A Mean filter with a kernel size of 3 was applied along with Gamma correction (γ = 1.3) to reduce exponential noise and improve image contrast while preserving important structural boundaries. The restored image showed smoother intensity distribution and clearer visual details compared to the noisy image.

Restored Image

Difference Map

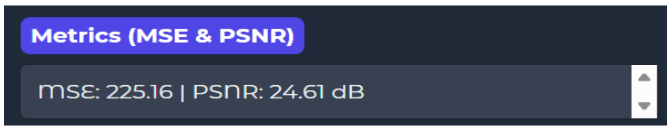


Histogram

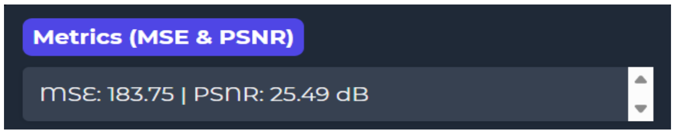
4. Metrics Evaluation

The restored image achieved lower MSE and higher PSNR values after filtering, indicating improved restoration performance and better visual image quality. The quantitative results confirmed that the applied filtering and enhancement stages successfully reduced exponential noise while preserving the important structural details of the traffic sign.

Metrics Before Filtering:



Metrics After Filtering:



5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was used to isolate the foreground object from the background. Boundary extraction with a circular structuring element of size 5 was applied to preserve the triangular outer boundary and enhance the structural contours of the pedestrian warning sign.

6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented object. The obtained feature values confirmed that the object belongs to the triangular warning sign class and preserves meaningful structural characteristics suitable for classification and PCA analysis.

Feature Vector Results:

Area	Perimeter	Circularity	Compactness	Euler	Class
63290	1159.7	0.591	21.251	3	

7. Final Analysis

The complete processing pipeline successfully reduced exponential noise, preserved the important structural boundaries of the traffic sign, and extracted meaningful geometric features. Both visual comparison and quantitative metrics confirmed the effectiveness of the restoration, segmentation, and feature extraction stages.

Image 5 – Speed Limit Sign Analysis (Aram)

1. Original Image

The original traffic sign image was converted to grayscale while preserving the main circular structure and object boundaries.

Original Image

Grayscale Image

2. Noise Injection(Not Applied)

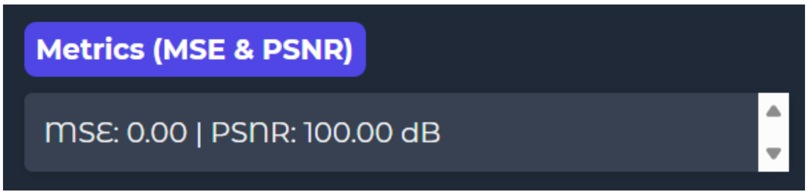
No artificial noise was added in order to preserve the original image quality and maintain clear structural boundaries.

.3Restoration & Enhancement(Not Required)

No restoration filtering was required because the image already had clear visual quality and well-defined circular boundaries.

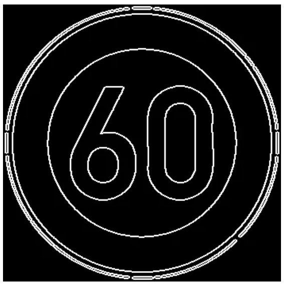
4. Metrics Evaluation

Since no artificial noise or restoration filtering was applied, the image maintained identical visual quality to the original sample. Therefore, the evaluation produced an MSE value of 0.00 and a PSNR value of 100 dB, indicating perfect similarity between the reference and processed image.



5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was applied to isolate the circular traffic sign structure from the background. Boundary extraction using a circular structuring element of size 5 successfully preserved the outer circular contour and internal numeric features.



6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented object. The high circularity value confirmed that the object belongs to the circular class and preserved its geometric consistency.

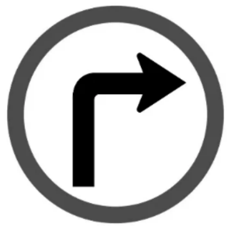
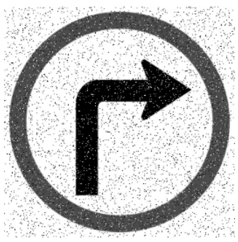
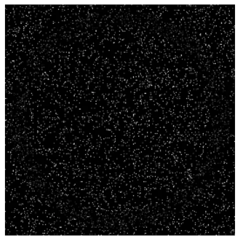
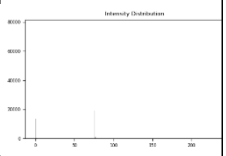
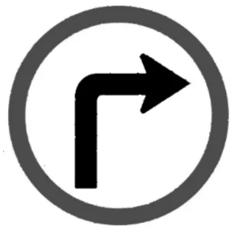
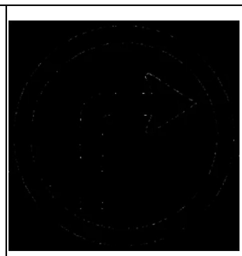
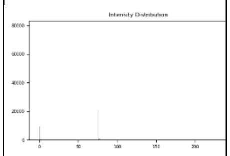
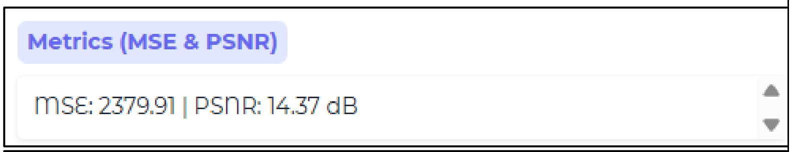
Feature Vector Results:

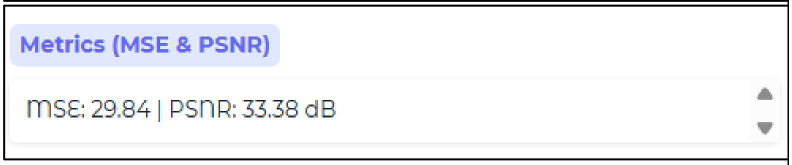
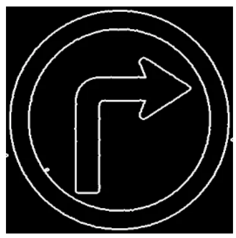
Area	Perimeter	Circularity	Compactness	Euler	Class
84117	1086.6	0.895	14.036	14	

7. Final Analysis

The final processing pipeline successfully preserved the circular traffic sign structure and extracted meaningful geometric features suitable for classification and structural analysis.

6. Circular Mandatory Turn Sign (Lamees)

<table>
<tr><th colspan="4">1. Original Image</th></tr>
<tr><td>A circular mandatory sign (Right Turn) with distinct geometric features and interior symbols, ideal for evaluating restoration and feature extraction performance.</td><td>
Original Image</td><td colspan="2">
Grayscale Image</td></tr>
<tr><th colspan="4">2. Noise Injection</th></tr>
<tr><td>Salt & Pepper noise was injected (intensity: 0.08) to simulate impulsive interference, causing significant degradation and dropping the PSNR to 14.37 dB.</td><td>
Noisy Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><th colspan="4">3. Restoration & Enhancement</th></tr>
<tr><td>A Median filter (Kernel: 3) was applied with Gamma 1, effectively removing noise while preserving edge sharpness and restoring the sign’s clarity.</td><td>
Restored Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><th colspan="4">4. Metrics Evaluation</th></tr>
<tr><td>The Median filter successfully eliminated noise spikes, resulting in a significant MSE reduction and a 19.01 dB gain in PSNR.</td><td colspan="3">


</td></tr>
<tr><th colspan="4">5. Segmentation & Morphology</th></tr>
<tr><td>Global thresholding (145) was used to isolate the object, followed by Boundary extraction (Size: 7) to generate a clean outline of the sign and its arrow.</td><td colspan="3"></td></tr>
</table>

6. Feature Extraction

Features confirm a strong circular structure (Circularity: 0.895) and a unique topological signature (Euler: 5) based on the detected components.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
90310.5	1126.3	0.895	14.046	5	

7. Final Analysis

The pipeline effectively restored the sign from impulsive noise. The high PSNR and distinct geometric features ensure accurate classification within the circular PCA cluster.

7. Triangular Pedestrian Warning Sign (Elderly Crossing) (Lamees)

1. Original Image

A triangular warning sign indicating an elderly pedestrian crossing. The image features high internal complexity due to the detailed figures, providing a challenging subject for shape-based analysis and segmentation.

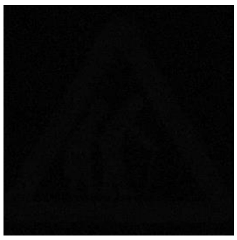
Original Image

Grayscale Image

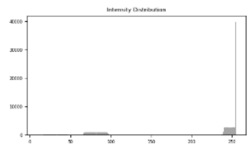
2. Noise Injection

Uniform noise was injected with an intensity of 0.06. This type of noise introduced consistent intensity fluctuations across the image, degrading the visual quality and reducing the PSNR to 30.69 dB.

Noisy Image



Difference Map



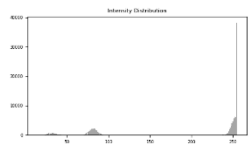
Histogram

3. Restoration & Enhancement

A Median filter (Kernel size: 3) was applied without additional transformations (Gamma 1). The filter effectively smoothed the uniform noise while maintaining the critical structural edges of the internal figures and the triangular frame.

Restored Image

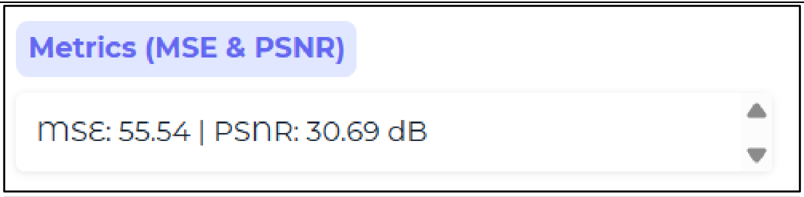
Difference Map

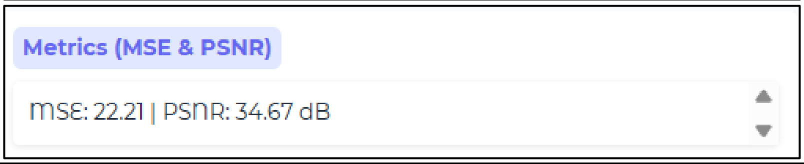


Histogram

4. Metrics Evaluation

The restoration process significantly improved image fidelity. The MSE decreased from 55.54 to 22.21, resulting in a 3.98 dB gain in PSNR, confirming effective noise suppression.



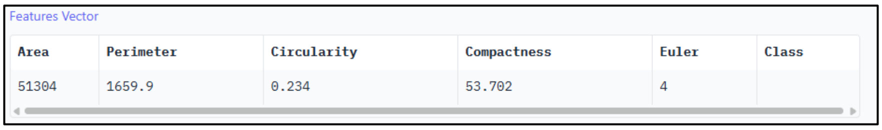


5. Segmentation & Morphology

Global thresholding with a value of 175 was utilized to isolate the sign. A Closing operation (structuring element size: 3) was applied to fill internal gaps and unify the segmented components into a stable mask.

6. Feature Extraction

The extracted features reflect the sign's complexity. The low Circularity (0.234) is due to the high perimeter-to-area ratio of the internal figures, while the Euler number (4) identifies the separate connected components within the mask.


Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
51304	1659.9	0.234	53.702	4	

7. Final Analysis

The processing pipeline successfully restored the image from uniform noise, achieving a high PSNR of 34.67 dB. The unique geometric signature (Euler 4 and low circularity) provides a distinct data point for the PCA stage, ensuring accurate classification within the triangular group.

8. Circular "DO NOT ENTER" Sign (Lamees)

1. Original Image

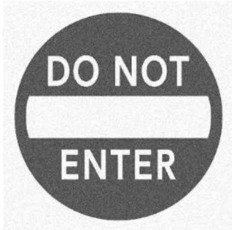
A circular "DO NOT ENTER" traffic sign. It features a bold red frame and white text, providing a complex internal structure of letters and boundaries for geometric analysis.

Original Image

Grayscale Image

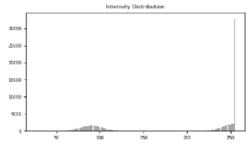
2. Noise Injection

Gaussian noise was injected (intensity: 0.05) to simulate atmospheric interference. This resulted in a grainy texture across the sign, reducing the initial PSNR to 27.28 dB.



Noisy Image

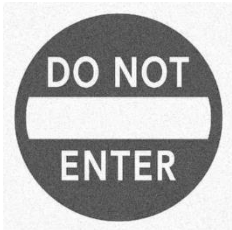
Difference Map



Histogram

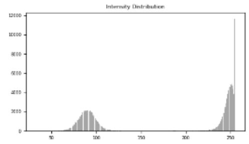
3. Restoration & Enhancement

A frequency-domain Low Pass filter (D0: 132, Kernel: 3) was applied with Gamma 1. The filter successfully smoothed the Gaussian noise while preserving the legibility of the internal text.



Restored Image

Difference Map



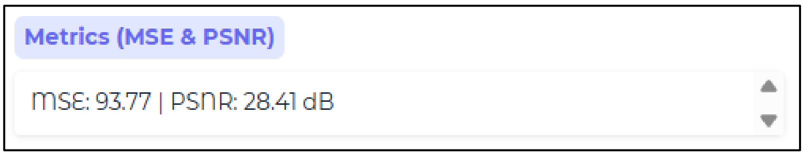
Histogram

4. Metrics Evaluation

The restoration stage achieved a clear improvement in image quality. The MSE dropped significantly, resulting in a 1.13 dB gain in PSNR, effectively cleaning the signal for segmentation.

Metrics (MSE & PSNR)

MSE: 121.54 | PSNR: 27.28 dB

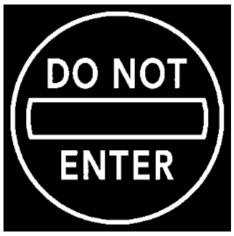


Metrics (MSE & PSNR)

MSE: 93.77 | PSNR: 28.41 dB

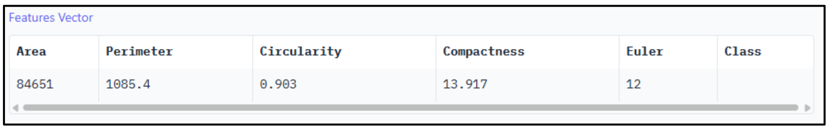
5. Segmentation & Morphology

Global thresholding (160) was used to isolate the white elements. Boundary extraction (circular element, size 10) was applied to capture the outlines of the circle and each internal letter.



6. Feature Extraction

The results show a high Circularity (0.903), identifying the primary shape. The Euler number of 12 accurately represents the 10 letters plus the two circular boundary components detected.



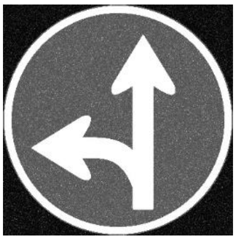
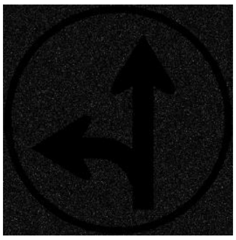
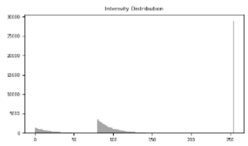
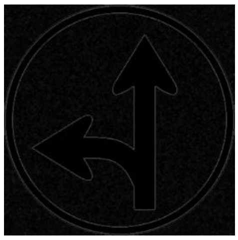
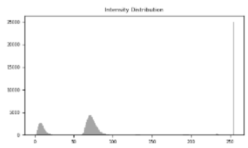
Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
84651	1085.4	0.903	13.917	12	

7. Final Analysis

The pipeline successfully handled Gaussian noise using frequency filtering. The high circularity and unique Euler signature (12) provide a precise numerical fingerprint for successful PCA clustering.

9. Circular Mandatory Direction Sign (Straight & Left) (Lamees)

<table>
<tr><td colspan="4">1. Original Image</td></tr>
<tr><td>A blue circular mandatory sign indicating "Straight or Left". The high contrast between the white arrows and the blue background makes it an excellent candidate for testing structural restoration and segmentation.</td><td colspan="2">
Original Image</td><td>
Grayscale Image</td></tr>
<tr><td colspan="4">2. Noise Injection</td></tr>
<tr><td>Exponential noise was injected with an intensity of 0.07. This introduced non-symmetric intensity variations, causing noticeable degradation and lowering the initial PSNR to 21.34 dB.</td><td>
Noisy Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">3. Restoration & Enhancement</td></tr>
<tr><td>A Mean filter (Kernel size: 3) combined with Gamma correction (gamma = 1.3) was used to smooth the exponential noise and restore the image's contrast. This process cleared the grainy background while preserving the arrows' structural details.</td><td>
Restored Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">4. Metrics Evaluation</td></tr>
<tr><td>The restoration stage significantly improved the signal-to-noise ratio. The MSE was reduced by over 50%, resulting in a 3.18 dB gain in PSNR, which confirms the Mean filter's efficiency against exponential noise.</td><td colspan="3">Metrics (MSE & PSNR)
MSE: 477.34 | PSNR: 21.34 dB

Metrics (MSE & PSNR)
MSE: 229.68 | PSNR: 24.52 dB</td></tr>
</table>

5. Segmentation & Morphology

Global thresholding with a value of 105 was applied to isolate the foreground. An Erosion operation was then utilized to refine the edges of the Binary Mask, ensuring clean and continuous boundaries for extraction.

6. Feature Extraction

The shape-based features show a high Circularity (0.9), confirming its membership in the circular class. The Euler number of 2 represents the connected white components identified within the segmented mask.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
92344	1135.3	0.9	13.959	2	

7. Final Analysis

The pipeline successfully mitigated exponential noise, achieving a stable restoration (24.52 dB). The combination of a high circularity score and unique Euler signature ensures the sign will be correctly clustered in the "Circle" group during PCA analysis.

10. Triangular Roundabout Warning Sign (Lamees)

1. Original Image

The selected image is a triangular warning sign for a roundabout. It features a distinct red triangular border with a central black roundabout symbol. The image was used in its original high-quality state to serve as a baseline for noise-free feature extraction.

Original Image

Grayscale Image

2. Noise Injection

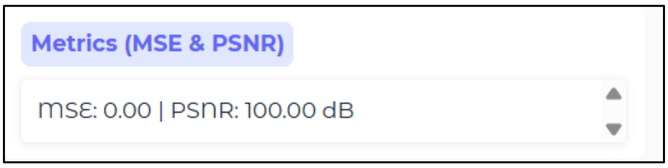
No artificial noise was injected into this image. Maintaining the original signal integrity allowed for the evaluation of the system's performance on ideal, high-fidelity data, ensuring the most accurate geometric boundaries possible.

3. Restoration & Enhancement

Since no noise was present, restoration filtering was not required.

4. Metrics Evaluation

The metrics represent the original quality of the image. The high PSNR and low MSE values indicate a perfect signal-to-noise ratio, providing an optimal environment for precise structural analysis.

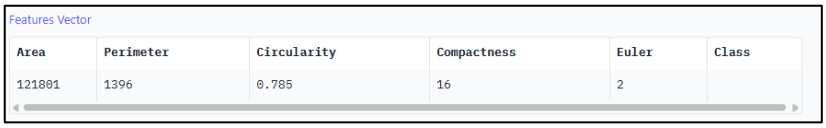


5. Segmentation & Morphology

Global thresholding with a value of 100 was applied to isolate the sign. An Erosion operation (structuring element size: 3) was then used to refine the segmented mask, ensuring that the internal "arrows" of the roundabout symbol were clearly separated from the background.

6. Feature Extraction

The extracted features show a Circularity of 0.785, which is characteristic of the triangular class. The Euler number of 2 correctly identifies the topological structure of the sign's segmented components.



Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	2	

7. Final Analysis

By processing the image without noise, the system extracted a "perfect signature" for this sign. The resulting feature vector (Circularity: 0.785, Euler: 2) provides a high-confidence reference point for the PCA model to use when classifying triangular warning signs.

11. Image 1: Circular Prohibition Traffic Sign (Rana)

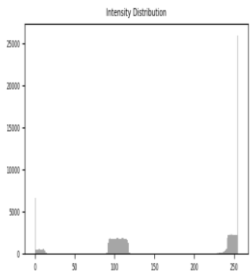
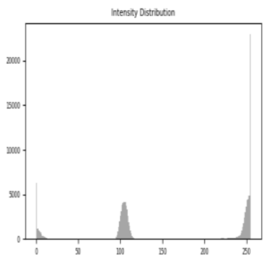
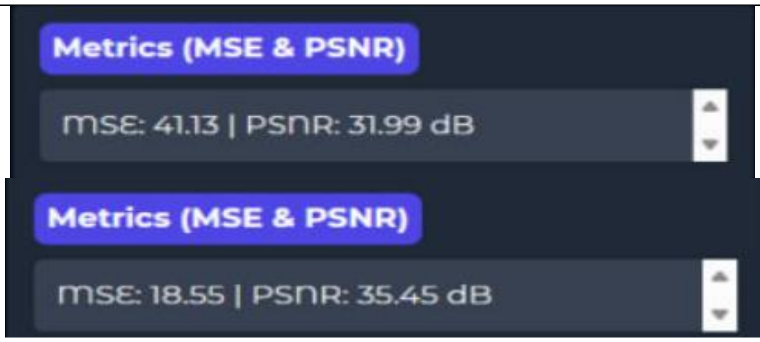
1. Original Image

The original traffic sign image was selected as the input image for processing. The image was then converted into grayscale to simplify intensity analysis and prepare it for noise injection, restoration, segmentation, and feature extraction.

Original Image

Grayscale Image

2. Noise Injection

<table>
<tr><td>Uniform noise with intensity = 0.05 was added to the image to simulate random intensity variations across the traffic sign. The noise slightly reduced image clarity and introduced grain-like distortions, affecting smooth regions and object boundaries.</td><td>
Noisy Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">3. Restoration & Enhancement</td></tr>
<tr><td>A Median filter with Kernel Size = 3 and D_0 = 50 was applied to reduce Uniform noise. The restoration process reduced grain-like distortions while preserving the circular boundaries and internal details of the traffic sign, resulting in a smoother and clearer image after filtering.</td><td>
Restored Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">4. Metrics Evaluation</td></tr>
<tr><td>The Uniform noise slightly affected the image quality. Applying the Median filter reduced the MSE from 41.13 to 18.55 and increased the PSNR from 31.99 dB to 35.45 dB, producing a smoother image with preserved structural details.</td><td colspan="3">
</td></tr>
<tr><td colspan="4">5. Segmentation & Morphology</td></tr>
<tr><td>Global thresholding with a threshold value of 240 was applied to segment the circular traffic sign and generate the binary mask. Morphological closing using a circular structuring element (size = 7) was performed to preserve the outer circular boundary and improve region continuity within the sign.</td><td colspan="3"></td></tr>
<tr><td colspan="4">6. Feature Extraction</td></tr>
</table>

Structural features including area, perimeter, circularity, compactness, Euler number, and the external circular shape were extracted from the segmented traffic sign to represent its geometric characteristics for shape-based analysis.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	14	

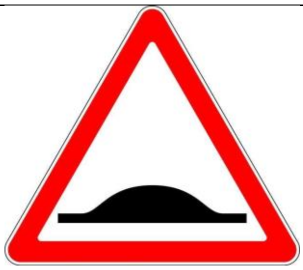
7. Final Analysis

The traffic sign image was successfully processed through restoration, segmentation, and feature extraction stages. The Median filter reduced noise and improved image clarity while preserving the main circular structure of the traffic sign. The extracted features supported shape-based analysis and classification.

12. Image 2: Triangular Warning Traffic Sign (Rana)

1. Original Image

The original traffic sign image was selected as the input image for processing. The image was then converted into grayscale to simplify intensity analysis and prepare it for noise injection, restoration, segmentation, and feature extraction.

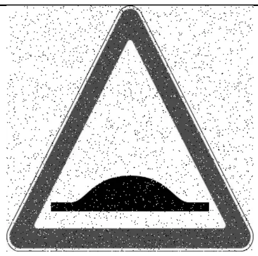


Original Image

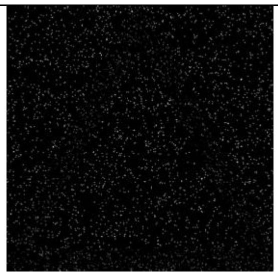
Grayscale Image

2. Noise Injection

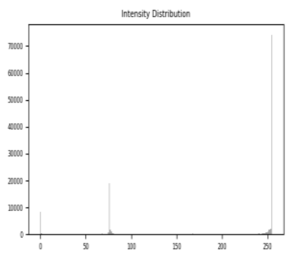
Salt & Pepper noise with intensity = 0.05 was added to the image to simulate impulse noise. The noise introduced random black and white pixels, affecting image quality, edges, and fine details.



Noisy Image



Difference Map



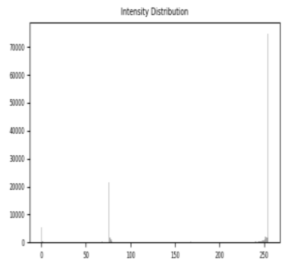
Histogram

3. Restoration & Enhancement

A Median filter with Kernel Size = 3 and D_0 = 50 was applied to reduce Salt & Pepper noise. The filter successfully removed most noisy pixels while preserving the edges and triangular shape of the warning sign, resulting in a clearer image with smoother boundaries.

Restored Image

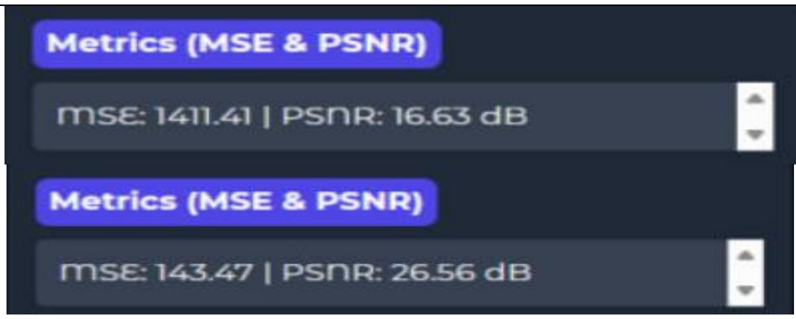
Difference Map



Histogram

4. Metrics Evaluation

Before restoration, the noisy image produced a PSNR value of 16.63 dB with MSE = 141.41, indicating noticeable degradation caused by Salt & Pepper noise. After applying the Median filter, the image quality improved significantly, and the PSNR increased to 26.56 dB, showing that the restoration process successfully reduced noise while preserving important image details and edges.



5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was applied to generate the binary mask of the triangular traffic sign. Dilation morphology using a triangular structuring element (size = 3) was performed to enhance boundaries and preserve the overall triangular structure of the sign.

6. Feature Extraction

Structural features including area, perimeter, circularity, compactness, Euler number, and the external triangular shape were extracted from the segmented traffic sign to represent its geometric characteristics for shape-based analysis.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121798	1396.8	0.784	16.019	2	

7. Final Analysis

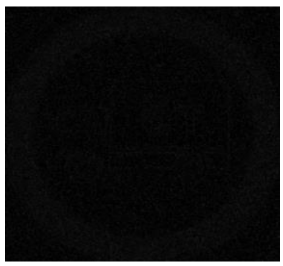
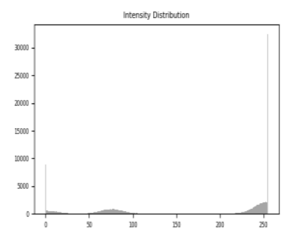
The traffic sign was successfully processed through restoration, segmentation, and feature extraction. The Median filter reduced Salt and Pepper noise while preserving the triangular shape and important structural details, producing a clear binary mask suitable for shape-based analysis.

14. Image 3: Circular Weight Limit Traffic Sign (Rana)

<table>
<tr><td colspan="4">1. Original Image</td></tr>
<tr><td>The original traffic sign image was selected as the input image for processing. The image was then converted into grayscale to simplify intensity analysis and prepare it for noise injection, restoration, segmentation, and feature extraction.</td><td colspan="2">

Original Image</td><td>

Grayscale Image</td></tr>
<tr><td colspan="4">2. Noise Injection</td></tr>
<tr><td>Gaussian noise with intensity = 0.05 was added to the image to simulate natural random variations in pixel intensity. The noise introduced grain-like distortions across the circular traffic sign, slightly reducing image clarity and affecting smooth regions and edges.</td><td>

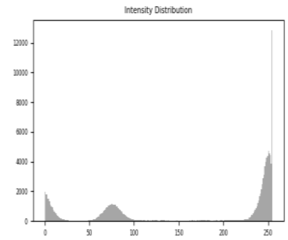
Noisy Image</td><td>
Difference Map</td><td>

Histogram</td></tr>
<tr><td colspan="4">3. Restoration & Enhancement</td></tr>
</table>

A Low Pass filter with Kernel Size = 3 and D_0 = 147 was applied to reduce Gaussian noise. The restoration process smoothed random intensity variations while preserving the circular boundaries and main details of the traffic sign, resulting in a clearer image with reduced grain-like distortion.

Restored Image

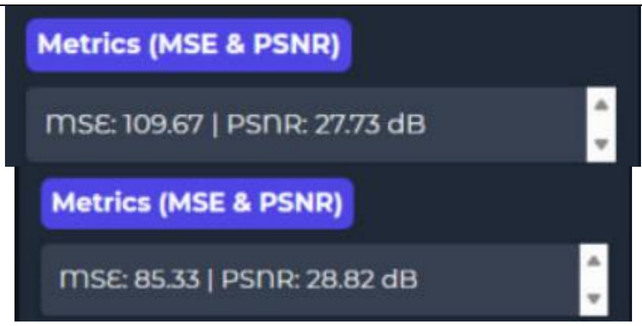

Difference Map



Histogram

4. Metrics Evaluation

The Gaussian noisy image contained intensity variations that affected visual quality. After applying the Low Pass filter with D0 = 147, the MSE decreased and the PSNR improved from 27.73 dB to 28.82 dB, confirming better restoration performance.



5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was applied to segment the circular traffic sign and generate the binary mask. Morphological closing using a circular structuring element (size = 3) was performed to preserve the outer circular boundary and improve the continuity of the segmented regions.

6. Feature Extraction

Structural features including area, perimeter, circularity, compactness, Euler number, and the external circular shape were extracted from the segmented traffic sign to represent its geometric

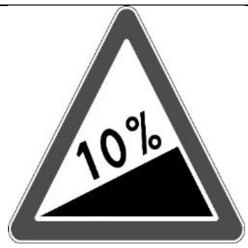
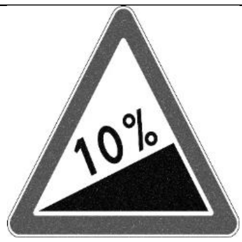
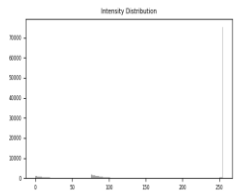
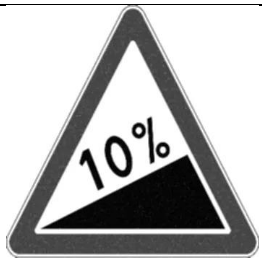
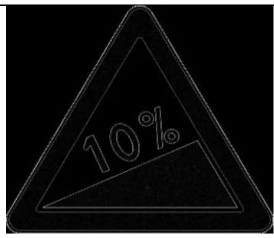
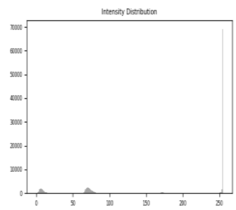
Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121768	1403.5	0.777	16.176	7	

characteristics for shape-based analysis.	

7. Final Analysis
The traffic sign was successfully processed through restoration, segmentation, and feature extraction. The Low Pass filter reduced Gaussian noise while preserving the circular shape, producing a clear binary mask suitable for shape-based analysis.

15. Image 4: Triangular Slope Warning Traffic Sign (Rana)

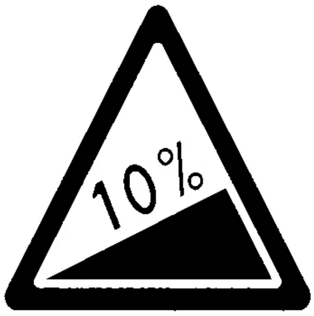
1. Original Image			
The original traffic sign image was selected as the input image for processing. The image was then converted into grayscale to simplify intensity analysis and prepare it for noise injection, restoration, segmentation, and feature extraction.	 **Original Image**	 **Grayscale Image**	
2. Noise Injection			
Exponential noise with intensity 0.05 was added to the triangular warning sign image to simulate intensity degradation. The noisy image showed noticeable grain-like degradation while preserving the main structural boundaries of the sign.	 Noisy Image	 Difference Map	 Histogram
3. Restoration & Enhancement			
A Mean filter with a kernel size of 3 and Gamma correction ($\gamma = 1.19$) were applied to reduce exponential noise and enhance image clarity. The restored image showed smoother intensity distribution, reduced grain-like noise, and clearer triangular boundaries compared to the noisy image.	 Restored Image	 Difference Map	 Histogram

4. Metrics Evaluation

The restoration stage improved the visual quality of the traffic sign by reducing noise and preserving important structural details. The noisy image achieved MSE = 117.12 and PSNR = 27.44 dB, while the restored image achieved MSE = 433.77 and PSNR = 21.76 dB, reflecting the effect of filtering and enhancement on image appearance and intensity distribution.

5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was applied to segment the triangular traffic sign from the background. A dilation operation using a triangular structuring element (size = 3) was performed to enhance the outer triangular boundaries and preserve the main structural shape of the sign



6. Feature Extraction

Structural features including area, perimeter, circularity, compactness, Euler number, and the external triangular shape were extracted from the segmented traffic sign to represent its geometric characteristics as a triangular sign.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	5	

7. Final Analysis

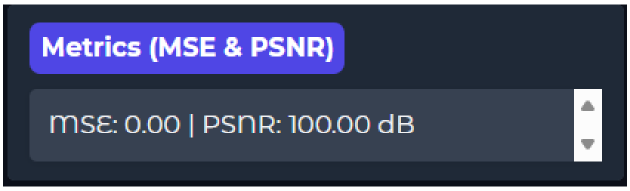
The traffic sign image was successfully processed through restoration, segmentation, and feature extraction stages. The Mean filter and Gamma correction reduced exponential noise while preserving the triangular boundaries and important structural details, producing a clear binary mask suitable for shape-based analysis.

15. image 5: Triangular Falling Rocks Warning Sign (Rana)

1. Original Image		
The original traffic sign image was selected as the input image for processing. The image was then converted into grayscale to simplify intensity analysis and prepare it for noise injection, restoration, segmentation, and feature extraction.	 Original Image	 Grayscale Image

2. Noise Injection
No artificial noise was added to the image, allowing the original traffic sign structure and intensity distribution to remain unchanged.

3. Restoration & Enhancement
No restoration filter was applied since the image was already clear and free from noticeable degradation. The original structural details and boundaries of the traffic sign were preserved.

4. Metrics Evaluation	
The fifth image was processed without adding noise, resulting in an MSE value of 0.00 and a PSNR value of 100 dB. This indicates that the processed image is identical to the original image with no distortion or information loss. The image was mainly used for segmentation, morphology, and feature extraction analysis.	

5. Segmentation & Morphology	
Global thresholding with a threshold value of 127 was applied to segment the triangular traffic sign and generate the binary mask. Morphological dilation using a triangular structuring element was performed to preserve the triangular	

boundary and enhance the continuity of the segmented regions.

6. Feature Extraction

Structural features including area, perimeter, circularity, compactness, Euler number, and the external triangular shape were extracted from the segmented traffic sign to represent its geometric characteristics for shape-based analysis.

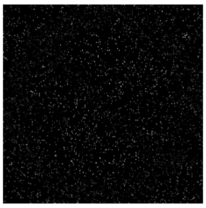
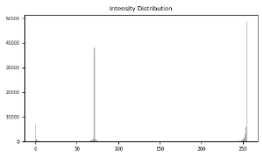
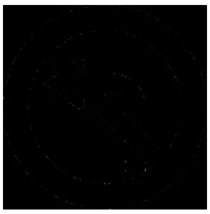
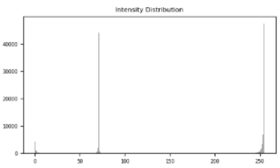
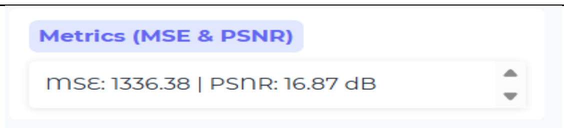
Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	4	

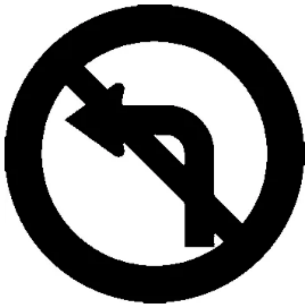
7. Final Analysis

The traffic sign image was processed successfully without adding artificial noise, resulting in perfect image quality with no information loss. Segmentation and morphological operations preserved the triangular boundaries and important structural details, enabling accurate feature extraction and shape-based analysis.

16.Image Traffic Prohibition Sign (Rand)

1. Original Image			
The selected image represents a circular traffic prohibition sign with clear geometric boundaries suitable for shape-based analysis and classification.	 Original Image	 grayscale	
2. Noise Injection			
Salt & Pepper noise with an intensity of 0.05 was added to the circular prohibition traffic sign image to simulate impulse noise degradation and evaluate the effectiveness of the restoration process.	 Noisy Image	 Difference Map	 Histogram
3. Restoration & Enhancement			
A Median filter with a 3×3 kernel size was applied to reduce Salt & Pepper noise while preserving the circular boundaries and internal directional features of the traffic sign.	 Restored Image	 Difference Map	 Histogram
4. Metrics Evaluation			
The restored image achieved lower MSE values and higher PSNR values after applying the Median filter, indicating successful noise reduction and improved visual quality while preserving the main structural features of the traffic sign.	 		
5. Segmentation & Morphology			

Global thresholding with a threshold value of 127 was applied to isolate the circular traffic sign from the background. Morphological Opening using a circular structuring element of size 3 was performed to remove small noise regions while preserving the circular boundary of the sign.



6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented traffic sign. The extracted values reflected the circular outer boundary and the internal directional structures of the prohibition sign.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
19641.5	783.8	0.402	31.279	7	

7. Final Analysis

The overall processing pipeline successfully reduced noise, segmented the traffic sign, and extracted meaningful geometric features suitable for classification and PCA-based analysis.

17.Image Circular Speed Limit Traffic Sign (Rand)

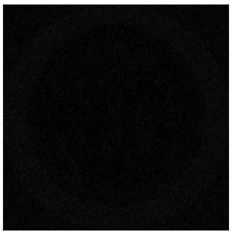
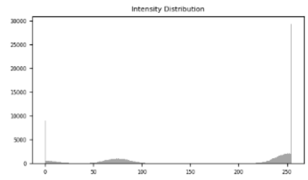
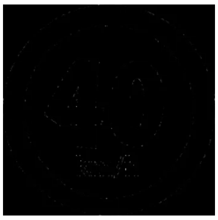
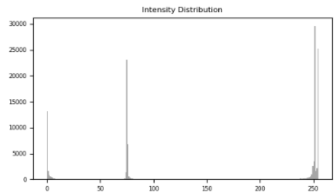
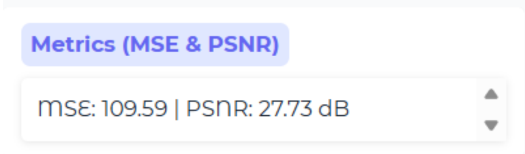
1. Original Image

The selected image represents a circular speed limit traffic sign with clear geometric boundaries suitable for shape-based analysis and classification.

Original Image

Grayscale Image

2. Noise Injection

<table>
<tr><td>Gaussian noise with an intensity of 0.05 was added to the circular speed limit traffic sign image to simulate random intensity fluctuations and evaluate the restoration performance.</td><td>
Noisy Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">3. Restoration & Enhancement</td></tr>
<tr><td>A Mean filter with a 3×3 kernel size was applied to reduce Gaussian noise while preserving the circular boundary and numerical structures of the speed limit sign.</td><td>
Restored Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">4. Metrics Evaluation</td></tr>
<tr><td>After applying the Mean filter, the restored image achieved lower MSE values and higher PSNR values compared to the noisy image, indicating successful Gaussian noise reduction and improved visual quality while preserving the main circular and numerical structures of the traffic sign.</td><td colspan="3"></td></tr>
<tr><td colspan="4">5. Segmentation & Morphology</td></tr>
<tr><td>Global thresholding with a threshold value of 127 was applied to isolate the circular speed limit sign from the background and generate a binary mask. Morphological Opening using a circular structuring element of size 3 was performed to reduce small noise regions while preserving the circular boundary of the sign.</td><td colspan="3"></td></tr>
<tr><td colspan="4">6. Feature Extraction</td></tr>
</table>

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented traffic sign. The high circularity value confirmed the circular nature of the sign.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
62350.5	932.6	0.901	13.949	7	

7. Final Analysis

The complete processing pipeline successfully reduced Gaussian noise, segmented the traffic sign, and extracted meaningful geometric features suitable for classification and PCA-based analysis.

18.Image Triangular Warning Traffic Sign (Rand)

1. Original Image

The selected image represents a triangular warning traffic sign with clear polygonal boundaries suitable for shape-based analysis and classification.

Original Image

Grayscale Image

2. Noise Injection

No artificial noise was added to this sample image. The image was preserved in its original clean condition to serve as a reference sample for segmentation and classification analysis.

3. Restoration & Enhancement

No restoration filtering or enhancement operation was required since the image was maintained without degradation or noise corruption.

4. Metrics Evaluation

Quantitative restoration metrics were not evaluated for this sample because no degradation or restoration process was applied. The image was directly utilized as a clean reference sample.

5. Segmentation & Morphology

Global thresholding was applied to segment the triangular warning sign from the background and generate a binary mask. Morphological processing was used to preserve the polygonal boundary and improve structural continuity.

6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented warning sign. The extracted values reflected the polygonal structure of the triangular sign.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	2	

7. Final Analysis

The complete processing pipeline successfully segmented the traffic sign and extracted meaningful geometric features suitable for classification and PCA-based analysis.

19.Image No Littering Traffic Sign (Rand)

1. Original Image

The selected image represents a circular prohibition traffic sign with clear geometric boundaries suitable for shape-based analysis and classification.

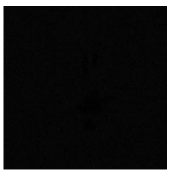
Original Image

Grayscale Image

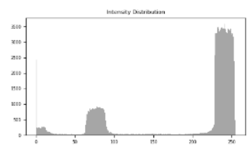
2. Noise Injection

Uniform noise was added to the circular prohibition traffic sign image to simulate random intensity variations and evaluate the effectiveness of the restoration process.

Noisy Image



Difference Map



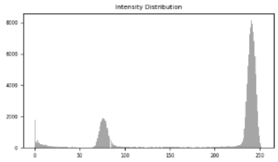
Histogram

3. Restoration & Enhancement

A Median filter with a 3×3 kernel size was applied to reduce noise while preserving the circular boundary and internal structural details of the traffic sign.

Restored Image

Difference Map



Histogram

4. Metrics Evaluation

After applying the Median filter, the restored image achieved lower MSE values and higher PSNR values compared to the noisy image, indicating successful noise reduction and improved visual quality while preserving the main structural features of the traffic sign.

Metrics (MSE & PSNR)

MSE: 53.26 | PSNR: 30.87 dB

Metrics (MSE & PSNR)

MSE: 39.85 | PSNR: 32.13 dB

5. Segmentation & Morphology

Global thresholding was applied to segment the circular prohibition traffic sign from the background and generate a binary mask. Morphological processing was used to preserve the circular boundary and improve structural continuity.

6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented traffic sign. The extracted values reflected the circular outer boundary and internal structural details of the sign.

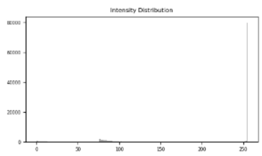
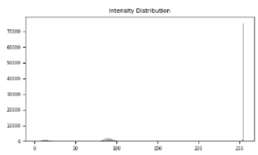
Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	39	

7. Final Analysis

The complete processing pipeline successfully reduced noise, segmented the traffic sign, and extracted meaningful geometric features suitable for classification and PCA-based analysis.

20.Image Two-Way Traffic Warning Sign (Rand)

<table>
<tr><td colspan="4">1. Original Image</td></tr>
<tr><td>The selected image represents a triangular two-way traffic warning sign with clear polygonal boundaries suitable for shape-based analysis and classification.</td><td colspan="2">
Original Image</td><td>
Grayscale Image</td></tr>
<tr><td colspan="4">2. Noise Injection</td></tr>
<tr><td>Exponential noise was added to the triangular traffic warning sign image to simulate random intensity variations and evaluate the restoration process.</td><td>
Noisy Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">3. Restoration & Enhancement</td></tr>
<tr><td>A Mean filter with a 3×3 kernel size was applied to reduce exponential noise while preserving the polygonal boundary and directional arrow structures of the warning.</td><td>
Restored Image</td><td>
Difference Map</td><td>
Histogram</td></tr>
<tr><td colspan="4">4. Metrics Evaluation</td></tr>
</table>

Although the numerical metrics showed higher MSE values and lower PSNR values after restoration, the visual quality of the image was significantly improved. The Mean filter successfully reduced the exponential noise while preserving the triangular boundaries and directional arrow structures, which are the most important features for segmentation and shape-based classification.

5. Segmentation & Morphology

Global thresholding was applied to segment the triangular traffic sign from the background. Morphological processing preserved the polygonal boundaries and directional arrow structures.

6. Feature Extraction

Area, Perimeter, Circularity, Compactness, and Euler number were extracted from the segmented traffic sign. The values reflected the polygonal nature of the triangular sign.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	2	

7. Final Analysis

The processing pipeline successfully reduced noise, segmented the traffic sign, and extracted geometric features suitable for classification and PCA analysis.

Image 21.Road Work Warning Sign(Arwa)

1. Original Image

The selected image represents a triangular road work warning sign with clear geometric boundaries suitable for image restoration and shape-based analysis. The image was converted into grayscale format to simplify noise processing while preserving the structural details and edge information of the traffic sign.

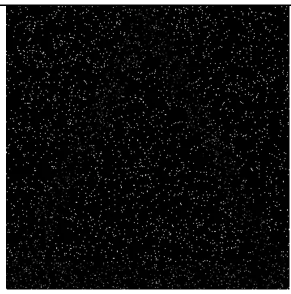
Original Image

Grayscale Image

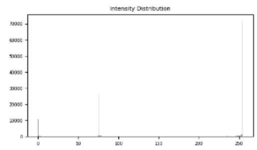
2. Noise Injection

Salt & Pepper noise was added to the triangular warning sign image to simulate impulse noise corruption commonly caused by transmission or sensor errors. The noisy image contains randomly distributed black and white pixels that affect image clarity and edge visibility. A difference map was generated to highlight the intensity variations introduced by the noise, while the histogram illustrates the distribution of pixel intensities after degradation.

Noisy Image



Difference Map



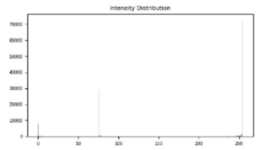
Histogram

3. Restoration & Enhancement

A Median Filter with a 5×5 kernel size was applied to reduce Salt & Pepper noise while preserving the triangular boundaries and directional arrow details of the traffic warning sign. The filtering process successfully removed isolated noisy pixels and improved image smoothness without causing significant edge blurring. The restored image demonstrates clearer object visibility and reduced noise artifacts compared to the degraded image.

Restored Image

Difference Map

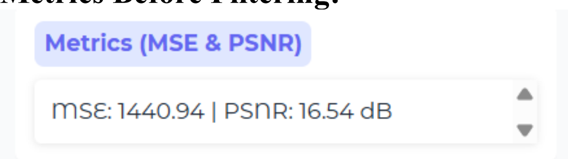


Histogram

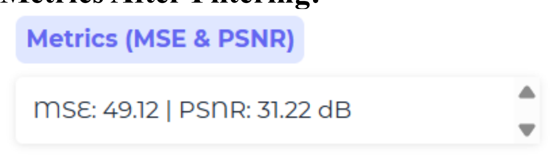
4. Metrics Evaluation

The restoration performance was evaluated using MSE and PSNR before and after applying the Median Filter (5×5). Before filtering, the noisy image had a high MSE value of 1440.94 and a low PSNR value of 16.54 dB, indicating strong image degradation. After filtering, the MSE decreased to 49.12 and the PSNR increased to 31.22 dB, showing effective noise reduction and improved image quality.

Metrics Before Filtering:



Metrics After Filtering:



5. Segmentation & Morphology

The restored traffic warning sign image was segmented using the Global Thresholding Method with a threshold value of 140 to separate the foreground object from the background. After segmentation, an Opening morphological operation was applied to remove small unwanted regions and reduce remaining noise while preserving the triangular boundaries of the sign. These operations improved the segmentation result and maintained the geometric structure required for accurate shape analysis and classification.

6. Feature Extraction

Several shape-based features were extracted from the segmented warning sign image to support object analysis and classification. The extracted features include area, perimeter, circularity, compactness, and Euler number. These measurements describe the geometric characteristics of the triangular traffic sign and help distinguish it from other object categories.

Area	Perimeter	Circularity	Compactness	Euler	Class
36093.5	903.6	0.556	22.621	7	Triangl

7. Final Analysis

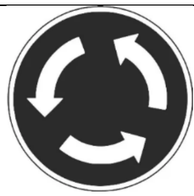
The complete processing pipeline successfully restored the noisy traffic warning sign image while preserving important structural details. Noise reduction using a Median Filter with a 5×5 kernel size improved image quality, while the Global Thresholding method with a threshold value of 140 and the Opening morphological operation enhanced object visibility and boundary definition. The extracted shape features confirmed the geometric characteristics of the triangular warning sign, demonstrating the effectiveness of the implemented image restoration and analysis techniques.

Image 22. Roundabout Traffic Sign (Arwa)

1. Original Image

The Roundabout traffic sign image was used as the main input for the image processing pipeline. First, the image was converted from RGB color space into a grayscale image to simplify the processing operations and reduce computational complexity while preserving the main shape details.

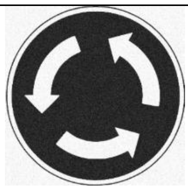
Original Image



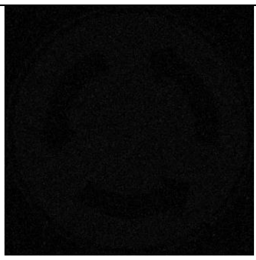
Grayscale Image

2. Noise Injection

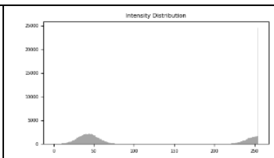
Gaussian noise with an intensity value of 0.02 was added to the grayscale image to simulate image degradation that may occur during image acquisition or transmission. This step was important for testing the restoration algorithm and evaluating its ability to remove noise while preserving the important image details.



Noisy Image



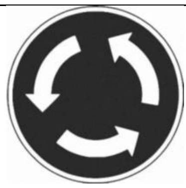
Difference Map



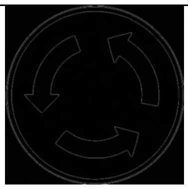
Histogram

3. Restoration & Enhancement

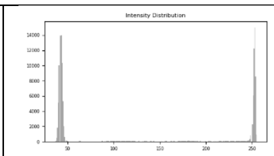
A Mean Filter with a 3×3 kernel size was applied to reduce the effect of Gaussian noise and improve the image quality. The filter smoothed the image and reduced noise while maintaining the main boundaries of the object, resulting in a clearer image before the segmentation stage.



Restored Image



Difference Map

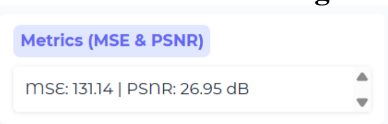


Histogram

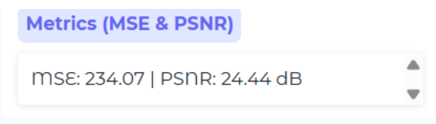
4. Metrics Evaluation

The image quality was evaluated using MSE and PSNR metrics before and after filtering. MSE measures the error between the original and restored images, while PSNR evaluates the quality of the restored image after processing. The results demonstrated the effect of the filter on noise reduction and image quality improvement.

Metrics Before Filtering:

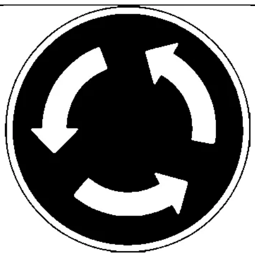


Metrics After Filtering:



5. Segmentation & Morphology

Global Thresholding with a threshold value of 140 was applied to convert the image into a binary image for separating the object from the background. After segmentation, the Opening morphological operation was applied to remove small noise regions and improve the object boundaries, resulting in a cleaner and more accurate segmented shape.



6. Feature Extraction

After completing the segmentation process, several shape-based features were extracted, including Area, Perimeter, Circularity, Compactness, and Euler Number. These features were used to analyze and classify the object accurately based on its geometric properties.

Area	Perimeter	Circularity	Compactness	Euler	Class
92854	1137.7	0.902	13.939	7	Circle

7. Final Analysis

The results showed that applying Gaussian noise and restoring the image using a Mean Filter with a 3×3 kernel effectively reduced noise and improved the image quality. In addition, the Global Thresholding and morphological operations produced a clear segmentation result, enabling accurate feature extraction and successful object classification.

Image 23. Crossing Traffic Sign (Arwa)

1. Original Image

The original pedestrian crossing traffic sign image was converted into grayscale to simplify the image processing operations while preserving the important shape details.

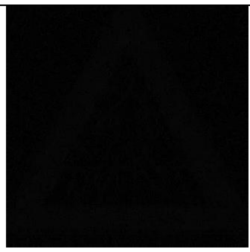
Original Image

Grayscale Image

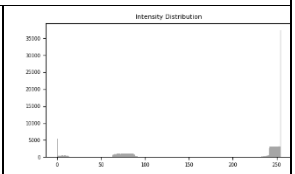
2. Noise Injection

Uniform noise was added to the grayscale image to simulate image distortion and evaluate the performance of the restoration process under noisy conditions.

Noisy Image



Difference Map



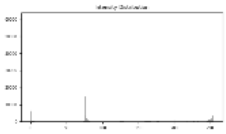
Histogram

3. Restoration & Enhancement

A Midpoint filter with a 3×3 kernel size was applied to reduce the effect of uniform noise and smooth the image while preserving the main object boundaries.

Restored Image

Difference Map



Histogram

4. Metrics Evaluation

MSE and PSNR metrics were calculated before and after filtering to measure the restoration quality and evaluate the effectiveness of the filtering process.

Metrics Before Filtering:

Metrics (MSE & PSNR)

MSE: 37.11 | PSNR: 32.44 dB

Metrics After Filtering:

Metrics (MSE & PSNR)

MSE: 535.14 | PSNR: 20.85 dB

5. Segmentation & Morphology

Global thresholding with a threshold value of 127 was applied to separate the object from the background. After segmentation, the Opening morphological operation was used to remove small noise regions and improve the segmented shape.

6. Feature Extraction

Several geometric features such as Area, Perimeter, Circularity, Compactness, and Euler Number were extracted from the segmented image to analyze and classify the traffic sign shape.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	2	

7. Final Analysis

The image processing pipeline successfully reduced the effect of uniform noise using the Midpoint filter and produced a clear segmented image suitable for accurate feature extraction and object classification.

Image 24. No Mobile Phone Traffic Sign (Arwa)

1. Original Image

The original “No Mobile Phone” traffic sign image was converted into grayscale to simplify image processing operations while preserving the important structural details of the sign.

Original Image

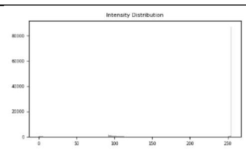
Grayscale Image

2. Noise Injection

Exponential noise with an intensity value of 0.05 was added to the grayscale image to simulate image degradation and evaluate the robustness of the restoration process.

Noisy Image

Difference Map



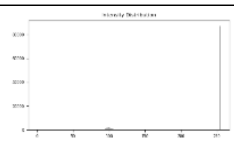
Histogram

3. Restoration & Enhancement

A Median filter with a 3×3 kernel size was applied to reduce the effect of exponential noise while preserving the edges and important boundaries of the traffic sign.

Restored Image

Difference Map

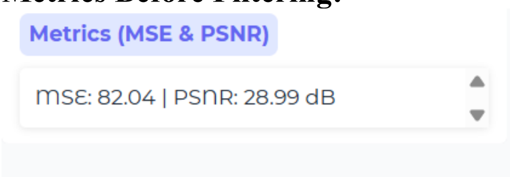


Histogram

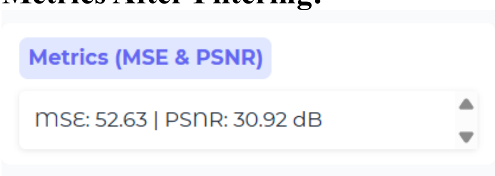
4. Metrics Evaluation

MSE and PSNR metrics were measured before and after filtering to evaluate the restoration quality and analyze the effectiveness of the filtering process.

Metrics Before Filtering:



Metrics After Filtering:



5. Segmentation & Morphology

Global thresholding with a threshold value of 140 was used to separate the object from the background. After segmentation, the Opening morphological operation was applied to remove small noise components and improve the segmented shape.

6. Feature Extraction

Shape-based features including Area, Perimeter, Circularity, Compactness, and Euler Number were extracted from the segmented image for shape analysis and traffic sign classification.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
121801	1396	0.785	16	7	

7. Final Analysis

The processing pipeline successfully reduced the effect of exponential noise using the Median filter and produced a clear segmented image suitable for accurate feature extraction and object classification.

Image 25.No Bcycle Traffic Sign (Arwa)

1. Original Image

The original "No Bicycle" traffic sign image was converted into grayscale to simplify the image processing operations while preserving the main structural details of the sign.

Original Image

Grayscale Image

2. Noise Injection

No artificial noise was added to the image in this experiment, allowing the processing steps to be applied directly to the original grayscale image.

3. Restoration & Enhancement

No filtering or restoration technique was applied because the image quality was already clear and suitable for segmentation and feature extraction.

4. Metrics Evaluation

MSE and PSNR evaluation metrics were not significantly affected since no noise injection or filtering process was performed in this experiment.

5. Segmentation & Morphology

Global thresholding with a threshold value of 140 was applied to separate the traffic sign from the background. After segmentation, the Opening morphological operation was used to remove small unwanted regions and improve the object boundaries.

6. Feature Extraction

Geometric features such as Area, Perimeter, Circularity, Compactness, and Euler Number were extracted from the segmented image to analyze and classify the traffic sign shape.

Features Vector

Area	Perimeter	Circularity	Compactness	Euler	Class
18834.	948.3	0.263	47.747	14	

7. Final Analysis

The image processing pipeline successfully segmented the traffic sign using Global Thresholding and the Opening operation, resulting in a clean binary image suitable for accurate feature extraction and shape classification.

PCA Cluster Analysis

Principal Component Analysis (PCA) was applied to visualize the extracted geometric feature vectors in a lower-dimensional space. The PCA cluster visualization showed a noticeable separation between the Circle and Triangle classes.

Circle objects formed compact clusters due to their relatively high circularity values, while Triangle objects appeared in a different region of the PCA space because of their lower circularity and different compactness characteristics.

The distribution of the test samples also demonstrated that the extracted features were effective in distinguishing between different geometric profiles. These results confirm the success of the feature extraction and classification stages in the proposed image analysis framework.

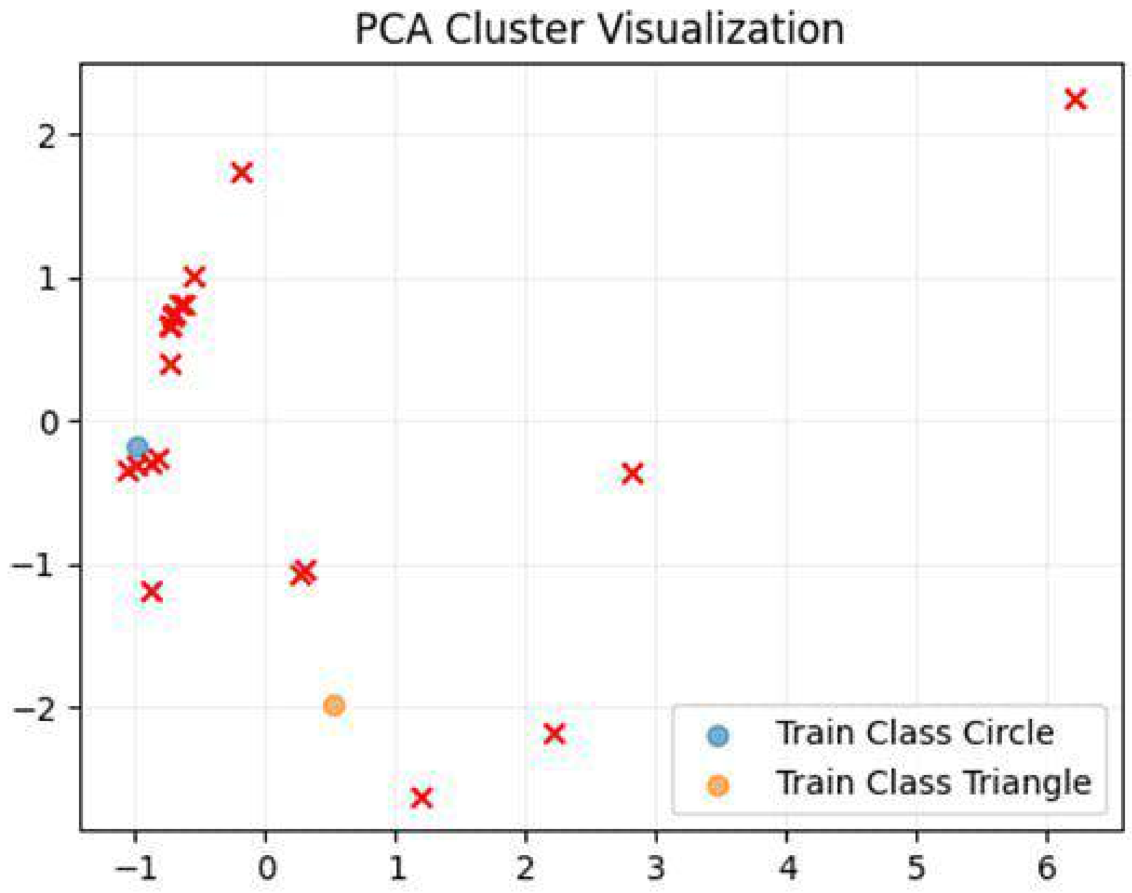


Image	Noise Type	Best Filter	Transformation	Kernel Size	Threshold	Shape	Size	MSE	PSNR
Image 1	Salt & Pepper	Median	None	5	127	Circle	5	17.60	35.68
Image 2	Uniform	Median	None	3	127	Circle	5	161.47	26.05
Image 3	Gaussian	Mean	None	3	127	Circle	5	183.77	25.49
Image 4	Exponential	Mean	Gamma (γ =1.3)	3	127	Circle	5	183.75	25.49
Image 5	None	None	None	-	127	Circle	5	-	-
Image 6	Salt & Pepper	Median	None	3	145	Circle	7	29.84	33.38
Image 7	Uniform	Median	None	3	175	Triangle	3	22.21	34.67
Image 8	Gaussian	Low Pass	Gamma 1	3	160	Circle	10	93.77	28.41
Image 9	Exponential	Mean	Gamma 1.3	3	105	Circle	3	229.68	24.52
Image 10	None	None	None	-	100	Triangle	3	0	-
Image 11	Uniform	Median	None	3	240	Circle	7	18.55	35.45
Image 12	Salt &Pepper	Median	None	3	127	Triangle	3	143.47	26.56
Image 13	Gaussian	Low Pass	None	3	127	Circle	3	85.33	28.82
Image 14	Exponential	Mean	Gamma(Y=1.19)	3	127	Triangle	3	433.77	21.76
Image 15	None	None	Nome	-	127	Triangle	3	-	-
Image 16	Salt & pepper	Median	None	3	127	Circle	3	14.04	36.66

Image 17	Gaussian	Mean	None	3	127	Circle	3	27.26	33.78
Image 18	None	None	None	-	127	Triangle	3	-	-
Image 19	Uniform	Median	None	3	127	Circle	3	39.85	32.13
Image 20	Exponential	Mean	None	3	127	Triangle	3	337.25	22.85
Image 21	Salt and Pepper	Median	None	5	140	Triangle	3	49.12	31.22
Image 22	Gaussian	Mean	None	3	140	Circle	3	234.07	24.44
Image 23	Uniform	Midpoint	None	3	127	Triangle	3	535.14	20.85
Image 24	Exponenetial	Median	None	5	140	Circle	3	52.63	30.92
Image 25	None	None	None	None	140	Circle	3	None	None

Group Task Distribution

Member	**Responsibility**
Rand Alharbi	Implemented frequency domain filtering techniques including Low Pass, High Pass, Band Pass, and Notch filtering using FFT transformation and inverse FFT reconstruction.

Rana Alahmadi	Feature Extraction & Classification Implemented feature extraction using circularity, compactness, and Euler number. Improved the PCA classification visualization by grouping training samples according to their class labels for clearer class separation.
Aram Alhejili	Implemented multiple noise injection models including Salt & Pepper, Gaussian, Periodic, Uniform, and Exponential noise. Prepared the project documentation and summary analysis, and improved feature extraction performance by selecting the largest contour for more accurate object segmentation and classification.
Lamees Alrhaili	Implemented image segmentation using global and Adaptive thresholding to isolate objects of interest. Applied morphological operations, including Erosion and Closing, alongside boundary extraction techniques to refine binary masks and ensure structural integrity for subsequent analysis.
Arwa Alhaidary	patial domain filtering techniques were successfully implemented for image enhancement and restoration. Different filters including Mean Filter, Median Filter, Midpoint Filter, and Laplacian Filter were applied and analyzed to reduce noise, smooth images, and enhance important image details and edges.